

Business License Data Profiling & Inference Report (DC, Chicago, NYC)

1. Overview

This report profiles three city business-license datasets and summarizes data quality issues at a column level. For each dataset, it documents inferred datatypes, maximum observed value widths, missingness, and key data issues that may affect analytics and downstream data engineering.

2. Washington, DC dataset

Field	Value
Row count	236,746
Column count	44
Primary license id	BBL_LICENSE_FACT_ID (0 duplicates detected in full scan)

Notable checks: ZIP format invalid rate ~0.49% (sample). Agent phone invalid rate ~7.96% (sample).

2.1 Column profiling (datatype, width, missingness, issues)

column	inferred_type	max_width	null_pct	issues
BILLING_ADDRESS	STRING	52	74.41	High missingness (74.41%).
BILLING_ADDRESS_CITY_STATE_ZIP	STRING	62	0.02	Some missing values (0.02%).
AGENT_ENTITY	STRING	92	47.87	Moderate missingness (47.87%).
AGENT_FIRST_NAME	STRING	30	58.58	High missingness (58.58%).
AGENT_LAST_NAME	STRING	42	58.58	High missingness (58.58%).
AGENT_MIDDLE_NAME	STRING	15	92.05	High missingness (92.05%).

AGENT_PHONE	STRING	15	38.09	Moderate missingness (38.09%). Standardize phone to 10-digit E.164-like format and validate.
SITE_ADDRESS	STRING	38	0.27	Some missing values (0.27%).
BBL_CITY_STATE_ZIP	STRING	42	0.19	Some missing values (0.19%).
BILLING_NAME	STRING	106	71.5	High missingness (71.5%).
CUSTOMER_NUMBER	BIGINT	16	0.0	No major issues detected from automated checks.
DC_ADDR_FLAG	STRING	1	0.0	No major issues detected from automated checks.
LICENSECATEGORY	STRING	29	0.0	No major issues detected from automated checks.
ENTITY_NAME	STRING	121	12.03	Moderate missingness (12.03%).
FIRST_NAME	STRING	37	78.52	High missingness (78.52%).
LAST_NAME	STRING	36	78.62	High missingness (78.62%).
DCS_ADD_DATE	STRING	0	100.0	High missingness (100.0%). Normalize to DATE/TIMESTAMP; validate for future/past bounds.
LICENSE_END_DATE	DATE/TIMESTAMP	22	4.68	Some missing values (4.68%). Normalize to

				DATE/TIMESTAMP; validate for future/past bounds.
LICENSE_ISSUE_DATE	DATE/TIMESTAMP	22	0.0	Normalize to DATE/TIMESTAMP; validate for future/past bounds.
LICENSE_START_DATE	DATE/TIMESTAMP	22	6.01	Some missing values (6.01%). Normalize to DATE/TIMESTAMP; validate for future/past bounds.
LICENSE_CATEGORY_TEXT	STRING	28	0.0	No major issues detected from automated checks.
LICENSE_CATEGORY_CODE	BIGINT	6	0.05	Some missing values (0.05%).
LICENSESTATUS	STRING	20	0.0	No major issues detected from automated checks.
PHONE_NUMBER	BIGINT	14	56.11	High missingness (56.11%). Standardize phone to 10-digit E.164-like format and validate.
OBJECTID	BIGINT	9	0.0	No major issues detected from automated checks.
CITY	STRING	30	0.24	Some missing values (0.24%).
STATE	STRING	2	0.71	Some missing values (0.71%).
ZIP	BIGINT	10	0.98	Some missing values (0.98%). Treat as STRING to preserve leading zeros.

BBL_LICENSE_FACT_ID	BIGINT	9	0.0	No major issues detected from automated checks.
DCS_LAST_MOD_DTTM	DATE/TIMESTAMP	22	0.0	Normalize to DATE/TIMESTAMP; validate for future/past bounds.
LATITUDE	DOUBLE	13	20.42	Moderate missingness (20.42%). Validate coordinate range and nullability.
LONGITUDE	DOUBLE	14	20.42	Moderate missingness (20.42%). Validate coordinate range and nullability.
XCOORD	DOUBLE	12	20.42	Moderate missingness (20.42%).
YCOORD	DOUBLE	12	20.42	Moderate missingness (20.42%).
ZIPCODE	STRING	0	100.0	High missingness (100.0%).
MARADDRESSREPOSITORYID	BIGINT	6	20.42	Moderate missingness (20.42%).
WARD	BIGINT	1	20.42	Moderate missingness (20.42%).
ANC	STRING	8	20.42	Moderate missingness (20.42%).
SMD	STRING	6	20.42	Moderate missingness (20.42%).
DISTRICT	STRING	7	20.42	Moderate missingness (20.42%).

PSA	BIGINT	3	20.42	Moderate missingness (20.42%).
NEIGHBORHOODCLUSTER	STRING	10	20.42	Moderate missingness (20.42%).
BUSINESSIMPROVEMENTDISTRICT	STRING	25	82.42	High missingness (82.42%).
MAINSTREET	STRING	0	100.0	High missingness (100.0%).

Notes: 'max_width' is the maximum observed string length (after reading values as text).
Inferred types are heuristic and should be confirmed with business rules.

2. Chicago dataset

Field	Value
Row count	1,133,370
Column count	34
Primary license id	LICENSE ID (0 duplicates detected in full scan)

Notable checks: ZIP format invalid rate ~0.06% (sample).

2.1 Column profiling (datatype, width, missingness, issues)

column	inferred_type	max_width	null_pct	issues
ID	STRING	16	0.0	No major issues detected from automated checks.
LICENSE ID	BIGINT	7	0.0	No major issues detected from automated checks.
ACCOUNT NUMBER	BIGINT	6	0.0	No major issues detected from automated checks.
SITE NUMBER	BIGINT	3	0.0	No major issues detected from automated checks.
LEGAL NAME	STRING	117	0.0	No major issues detected from automated checks.
DOING	STRING	128	0.03	Some missing

BUSINESS AS NAME				values (0.03%).
ADDRESS	STRING	92	0.0	No major issues detected from automated checks.
CITY	STRING	30	0.0	No major issues detected from automated checks.
STATE	STRING	2	0.0	No major issues detected from automated checks.
ZIP CODE	BIGINT	5	0.03	Some missing values (0.03%). Treat as STRING to preserve leading zeros.
WARD	BIGINT	2	7.51	Some missing values (7.51%).
PRECINCT	BIGINT	4	7.66	Some missing values (7.66%).
WARD PRECINCT	STRING	6	7.49	Some missing values (7.49%).
POLICE DISTRICT	BIGINT	3	7.56	Some missing values (7.56%).
LICENSE CODE	BIGINT	4	0.0	No major issues detected from automated checks.
LICENSE DESCRIPTION	STRING	60	0.0	No major issues detected from automated checks.
BUSINESS ACTIVITY ID	STRING	81	48.07	Moderate missingness (48.07%).
BUSINESS ACTIVITY	STRING	735	48.07	Moderate missingness (48.07%). Very long text values; consider truncation or separate notes field.
LICENSE	BIGINT	7	0.0	No major issues

NUMBER				detected from automated checks.
APPLICATION TYPE	STRING	6	0.0	No major issues detected from automated checks.
APPLICATION CREATED DATE	DATE/TIMESTAMP	10	76.71	High missingness (76.71%). Normalize to DATE/TIMESTAMP; validate for future/past bounds.
APPLICATION REQUIREMENTS COMPLETE	STRING	10	1.62	Some missing values (1.62%).
PAYMENT DATE	DATE/TIMESTAMP	10	1.98	Some missing values (1.98%). Normalize to DATE/TIMESTAMP; validate for future/past bounds.
CONDITIONAL APPROVAL	STRING	1	0.0	No major issues detected from automated checks.
LICENSE TERM START DATE	DATE/TIMESTAMP	10	0.24	Some missing values (0.24%). Normalize to DATE/TIMESTAMP; validate for future/past bounds.
LICENSE TERM EXPIRATION DATE	DATE/TIMESTAMP	10	0.01	Some missing values (0.01%). Normalize to DATE/TIMESTAMP; validate for future/past bounds.
LICENSE APPROVED FOR ISSUANCE	STRING	10	4.67	Some missing values (4.67%).
DATE ISSUED	DATE/	10	0.0	Normalize to

	TIMESTAMP			DATE/TIMESTAMP; validate for future/past bounds.
LICENSE STATUS	STRING	3	0.0	No major issues detected from automated checks.
LICENSE STATUS CHANGE DATE	DATE/ TIMESTAMP	10	93.3	High missingness (93.3%). Normalize to DATE/TIMESTAMP; validate for future/past bounds.
SSA	BIGINT	2	74.0	High missingness (74.0%).
LATITUDE	DOUBLE	12	7.73	Some missing values (7.73%). Validate coordinate range and nullability.
LONGITUDE	DOUBLE	13	7.73	Some missing values (7.73%). Validate coordinate range and nullability.
LOCATION	STRING	40	7.73	Some missing values (7.73%).

Notes: 'max_width' is the maximum observed string length (after reading values as text).

Inferred types are heuristic and should be confirmed with business rules.

2. New York City dataset

Field	Value
Row count	281,413
Column count	27
Primary license id	DCA License Number (2 duplicates detected in full scan)

Notable checks: ZIP format invalid rate ~0.09% (sample). Contact phone invalid rate ~0.88% (sample).

2.1 Column profiling (datatype, width, missingness, issues)

column	inferred_type	max_width	null_pct	issues
DCA License Number	STRING	16	0.0	No major issues detected from automated checks.
License Type	STRING	10	0.0	No major issues detected from automated checks.
License Expiration Date	DATE/ TIMESTAMP	10	7.49	Some missing values (7.49%). Normalize to DATE/TIMESTAMP; validate for future/past bounds.
License Status	STRING	8	0.0	No major issues detected from automated checks.
License Creation Date	DATE/ TIMESTAMP	10	0.0	Normalize to DATE/TIMESTAMP; validate for future/past bounds.
Industry	STRING	30	0.0	No major issues detected from automated checks.
Business Name	STRING	91	0.02	Some missing values (0.02%).
Business Name 2	STRING	93	81.98	High missingness (81.98%).
Address Building	STRING	20	33.96	Moderate missingness (33.96%).
Address Street Name	STRING	40	33.57	Moderate missingness (33.57%).
Secondary Address Street Name	STRING	61	99.04	High missingness (99.04%).
Address City	STRING	30	0.29	Some missing values (0.29%).
Address State	STRING	30	0.36	Some missing

				values (0.36%).
Address ZIP	BIGINT	5	0.31	Some missing values (0.31%). Treat as STRING to preserve leading zeros.
Contact Phone Number	STRING	24	36.17	Moderate missingness (36.17%). Standardize phone to 10-digit E.164-like format and validate.
Address Borough	STRING	13	34.12	Moderate missingness (34.12%).
Borough Code	BIGINT	1	39.37	Moderate missingness (39.37%).
Community Board	BIGINT	3	41.04	Moderate missingness (41.04%).
Council District	BIGINT	3	40.87	Moderate missingness (40.87%).
BIN	BIGINT	19	41.63	Moderate missingness (41.63%).
BBL	BIGINT	11	41.63	Moderate missingness (41.63%).
NTA	STRING	4	51.15	High missingness (51.15%).
Census Tract	BIGINT	6	51.15	High missingness (51.15%).
Detail	STRING	255	85.71	High missingness (85.71%). Very long text values; consider truncation or separate notes field.
Longitude	DOUBLE	18	37.27	Moderate

				missingness (37.27%). Validate coordinate range and nullability.
Latitude	DOUBLE	18	37.27	Moderate missingness (37.27%). Validate coordinate range and nullability.
Location	STRING	40	37.27	Moderate missingness (37.27%).

Notes: 'max_width' is the maximum observed string length (after reading values as text).
Inferred types are heuristic and should be confirmed with business rules.

3. Cross-dataset common issues and recommendations

Common issues observed across datasets include: inconsistent representations of status, missing contact information (especially in Chicago extract), varying industry/license-type taxonomies, and partial geographic attributes. Recommended remediation steps:

- Standardize all identifiers (license numbers, account/customer numbers, ZIP) to STRING.
- Normalize status into a 2-level status_group (Active/Inactive) plus a detailed status_name.
- Create a conformed industry dimension; optionally map raw industry/license text into broader industry_group buckets.
- Standardize addresses: trim whitespace, handle 'Unknown', and extract street_name consistently.
- Validate latitude/longitude ranges and treat out-of-range values as null/quarantine.
- Apply DQ rules with Databricks DQX and route failing records into a quarantine table for review.